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11BR · Topic 6: Security, Privacy and Data Integrity

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Total marks: 56

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 56 marks

Question 1 · May/June 2023 Paper 13, original Q4 (d) [5 marks]

Question	Answer	Marks
1(a)(i)	**1 mark** each to **max 2** • Length check: the registration number must be **6 characters** long • Format check: the registration number must be in the format **letter-digit-digit-digit-letter-letter** • Type check: the registration number must be **alphanumeric**	**2**
1(a)(ii)	**1 mark** each • Visual check: **Manually** compare the registration number entered with the source document • Double entry: Enter the registration number twice and **the computer compares** to check they are the same	**2**
1(a)(iii)	The registration number on the original document might be in the correct format but may be the incorrect registration number for that car.	**1**

Question 2 · May/June 2024 Paper 12, original Q3 (a) [6 marks]

Question	Answer	Marks

2(a)(i)	**1 mark** for security measure **1 mark** each to **max 2** for how the chosen measure works: • Firewall • Checks **incoming** connections • ... against criteria • Blocks data from entering specific ports • Blocks data that does not meet whitelist that meets blacklist • Proxy server • Prevents devices accessing the web server directly • Intercepts any requests • Forwards the request using its own IP address • Screens returning data before sending it to the user	**3**
2(a)(ii)	**1 mark** for security measure **1 mark** each to **max 2** for description of the chosen measure: • Encryption • Encodes/scrambles data • ... so if it is intercepted it cannot be **understood** • Algorithm/key is required to decode the data	**3**

Question 3 · May/June 2026 Paper 12, original Q2 (a),(b),(c) [13 marks]

Question	Answer	Marks	Guidance
3	Examiners must award **whole marks** (not half marks, or other fractions).		

3(a)	**1 mark** for method **1 mark** for correct matching description (* **2**) • Visual check • The entered data is **compared with the original** by a person • Double entry • Enter the data twice and the computer automatically **compares** if they match	**4**	Note: Question asks for methods of data **verification** not validation Accept examples of visual checks, such as printing a copy and comparing the printout **to the original** Data as intended is NE. It is the comparison to the original that is important
3(b)	**1 mark** for two correct names of methods **1 mark** for each correct matching use to **max 2** • Presence check • The Date of Birth (DoB) field cannot be left blank • Range check • DoB must be after one date **and** before another date // between specific dates – must include both ends • Limit check • DoB must be after or before a specific date – must be only one end • Format check • DoB must be dd/mm/yyyy • Length check • DoB must be **exactly** 10 characters • Type check • The data type must be DATE or DATE/TIME	**3**	Note: Question asks for methods of data **validation** not verification Mark the first answer in each box The use must match the example given in the question 05/07/2010 Do not accept range and limit as two different checks Take care that answers do not just repeat the method, must be applied in some way. The use must match the method If no methods are given read the uses which must be applied to the DoB. For format check accept mm/dd/yyyy

3(c)(i)	**1 mark** for correct description of a virus • **Description:** **malicious** software that replicates on the computer and can fill storage or delete/corrupt/change data - cause harm is NE **1 mark** each for a correct method and matching description, **2 marks** • **Method:** Anti-virus software // anti-malware software • **Description:** The software scans files/downloads/device and compares against database of known viruses, and quarantines/deletes/alerts the user if a match found	**3**	The description of the method must match the method chosen
3(c)(ii)	**1 mark** for correct description of a hacker • **Description:** person who gains **unauthorised access** to a computer system **1 mark** each for a correct method and matching description, **2 marks** For example: • **Method:** Firewall • **Description:** The firewall monitors all incoming and outgoing data compares it to stored criteria and blocks transmissions that are not allowed • **Method: Multi-factor** authentication • **Description:** A second form of verification such as biometrics or use of an authenticator app is required for logging in • **Method:** Strong passwords • **Description:** • The password contains a mixture of different types of characters and is at least a set number of characters and is not something connected to the individual such as a date of birth • **Method:** Encryption • **Description:** Scramble/encode the data/turn into ciphertext so that even if it is accessed it will not be **understood/useable** without the decryption key	**3**	The description of the method must match the method chosen Just malicious intent is NE for unauthorised access Accept two-factor for multi-factor If firewall is the candidate's answer for **question 2(c)(i)** ensure it is not repeated here Just authentication is NE for multi-factor authentication Map biometrics/use of biometric password as the method to strong password **Description:** Uses features of an individual that are unique and which are almost impossible to duplicate

Total: 56 marks

Question 4 · May/June 2026 Paper 13, original Q1 [5 marks]

Question	Answer	Marks	Guidance
4	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.		
	4 0	1	0
	4 1	1	1

Question 5 · Oct/Nov 2023 Paper 12, original Q5 (a),(b),(c) [6 marks]

Question	Answer	Marks
5(a)	**1 mark** for: Either • Ensuring data can only be accessed by / disclosed to authorised persons Or • Ensuring data cannot be accessed by / disclosed to unauthorised persons	1
5(b)	**1 mark** for each bullet point (**max 1**) • Ensuring the accuracy / completeness / consistency of data (during / after processing) • Ensuring the data is up to date	1

5(c)	**1 mark** for each bullet point. Phishing email (**max 2**) • The email pretends to be from an official body • ... persuading individuals to disclose private information // by example such as bank details • ... or requesting authentication by redirecting to an unofficial/unauthorised website // inviting a user to click a link Spyware (**max 2**) • Malware downloaded **without the user's knowledge** • ... which secretly records the user's actions / keystrokes on the computer • ... and sends logs of the actions to a third party	**4**

Question 6 · Oct/Nov 2023 Paper 13, original Q3 (c) [4 marks]

Question	Answer	Marks

6(a)	**1 mark** for each bullet point. • The administrator completes a visual check / checks by eye • …that the tutor identifier input matches the tutor identifier on the **original** document • Double entry check // the administrator (or a second person) enters the number a second time • …and the **system** compares it with the first entry	**4**

Question 7 · Oct/Nov 2023 Paper 13, original Q8 (a),(b),(c) [7 marks]

Question	Answer	Marks
7(a)	**1 mark** for each bullet point (**max 2**) • Validation // a validation method named or described • …protects the data by ensuring that the data is reasonable / sensible and within specified bounds	**2**

7(b)	**1 mark** for difference **1 mark** for similarity Difference: • Pharming is malicious code that redirects to a **fake website**. Phishing uses an **email** to prompt user action. • Pharming is **automatic**. Phishing requires **user action**. Similarity: • Both try to obtain financial or personal information • Both are a false representation of an official organisation, e.g. a bank • Both make use of fake websites	**2**

7(c)	**1 mark** for each bullet point (**max 3**). • Download programs from reputable websites / sources • …as these are less likely to contain malware • Backup / archive computer systems • …so they can be restored in case of data loss from malware program installation • **Install and run** anti-malware program • …so that regular scans can be made for known malware • …and if malware is found it can be quarantined / removed • …and computer's anti-malware definitions are regularly updated • Using a firewall to block unused ports • …so that malware cannot enter the computer system • Deny administrator privileges to everyday users • …so that malware cannot be downloaded by everyday users • Avoid the use of / access to removable devices • …so that malware cannot be installed from these devices	**3**

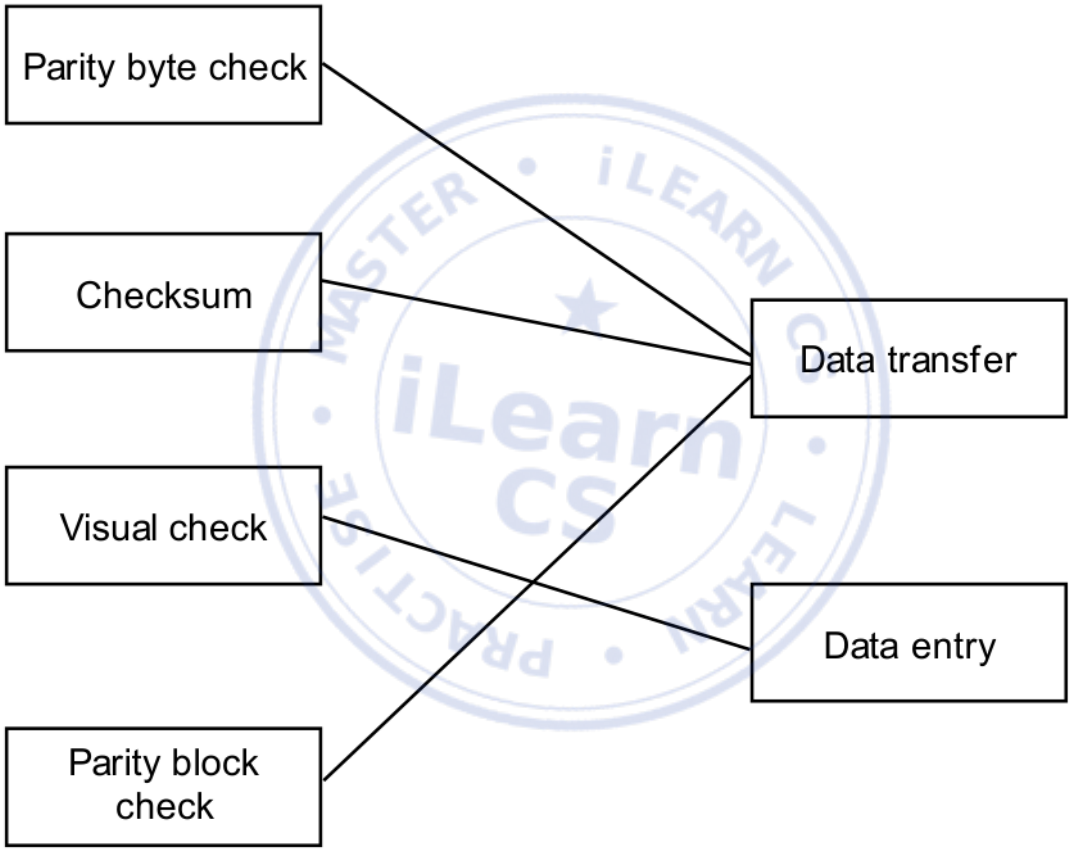
Question 8 · Oct/Nov 2024 Paper 12, original Q4 (c) [3 marks]

Question	Answer	Marks

8(a)	**1 mark** for the security method. **2 marks** for explanation Security method: Encryption Explanation • File contents are converted to cipher text • If intercepted the data cannot be understood without the decryption key	**3**

Question 9 · Oct/Nov 2025 Paper 12, original Q1 [2 marks]

Question	Answer	Marks

9	**1 mark** for 2 or 3 correctly connected verification method boxes, **2 marks** for all 4 correct **Verification Method** **Use**  	**2**

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Total: 56 marks

Question 10 · Oct/Nov 2025 Paper 12, original Q6 (c) [2 marks]

Question	Answer	Marks
10(a)(i)	**1 mark** for correct answer parity bit ↓ 0 1 0 1 1 1 0 **0**	1
10(a)(ii)	**1 mark** for correct answer parity bit ↓ 1 0 1 1 0 1 1 1 0 1 1 1 0 0 0 0 0 0 0 1 1 0 1 1 0 1 1 1 0 1 0 0 parity byte→ 1 0 1 0 0 0 0 0 [1]	1

Question 11 · Oct/Nov 2025 Paper 13, original Q7 (e) [3 marks]

Question	Answer	Marks

11(a)	**1 mark** for each validation check correctly identified, **max 3** marks Range Check Existence Check Format Check	**3**

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